

GUILHERME I. GONÇALVES

Senior Full-Stack Software Engineer

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EXPERIENCE

Senior Full-Stack Software Engineer

Siteimprove

Feb 2022 – Present Remote – Brazil

- Contributes to large-scale, cloud-based systems at **Siteimprove**, a global SaaS platform for digital accessibility and content quality used by enterprise organizations worldwide.
- Designs, implements, and maintains distributed back-end services in **Java (Spring Boot)**, **C# (ASP.NET)**, and **Node.js (Express)**, exposing **RESTful APIs** consumed by multiple client applications.
- Develops front-end applications and shared libraries using **JavaScript/TypeScript** and **React**, including the **Fancy Design System** component library with accessibility compliance (**WAI-ARIA**).
- Contributes to cloud-native systems deployed on **AWS**, working with both **SQL (Oracle)** and **NoSQL (DynamoDB)** stores, together with CloudFront, ElastiCache, and Redis.
- Maintains services on **EKS** in **FedRAMP**-compliant environments and develops against cluster-hosted services using port-forwarding.
- Integrates **AI-based APIs** into product features, enabling data-driven insights and enhancing digital experiences for customers.
- Implements authentication and authorization mechanisms based on **OAuth 2.0**, **OpenID Connect**, and **SAML 2.0** for distributed services and web applications.

Co-founder & Founding Engineer

Orkast

Mar 2026 – Present Remote – Brazil

- Co-founded Orkast, a multi-tenant **SaaS B2B** platform (ERP + CRM + AI) for Portuguese SMEs, and acts as sole engineer owning the full product lifecycle from requirements to operations.
- Designed and implemented the back-end in **Node.js (NestJS)** and **TypeScript**, exposing **RESTful APIs** backed by **PostgreSQL** through an **ORM** layer (Prisma), with row-level multi-tenant isolation, **JWT**-based authentication, and role-/module-based authorization.
- Developed the front-end application in **React** and **TypeScript** (Vite, MUI, TanStack Query), including Kanban boards, financial dashboards, and reporting interfaces used by tenant organizations.
- Designed and operates the entire cloud infrastructure on **AWS** using **Terraform** (infrastructure as code), including **Aurora PostgreSQL Serverless**, **ECS Fargate**, ALB with HTTPS, ACM, ECR, and S3, with container-based deployments through **Docker**.
- Integrated generative-AI capabilities via the **Anthropic Claude API** for business diagnostics, CSV column inference in bank reconciliation, and natural-language reporting over tenant data.

User & Scientific Researcher

CERN – European Organization for Nuclear Research

Mar 2019 – Present Geneva, Switzerland

- Worked on-site from 2019 to 2020 designing and implementing advanced **energy reconstruction algorithms** in **C++** and **Python** for the ATLAS Tile Calorimeter, contributing to the **Athena** software framework (a large-scale open-source platform) and targeting high pile-up conditions at the **Large Hadron Collider**.

EDUCATION

Ph.D. in Computational Modelling

Rio de Janeiro State University

Mar 2021 – Feb 2026

- Research in **Python** and **C++** on energy reconstruction algorithms for **ATLAS** at **CERN**.

Master's in Computational Modelling

Rio de Janeiro State University

Jan 2018 – Apr 2020

Bachelor's in Computer Engineering

Rio de Janeiro State University

Jan 2012 – Aug 2017

SKILLS

Programming Languages

Python C++ Java JavaScript
TypeScript C# C Ruby Shell

Software Engineering

Distributed Systems RESTful APIs
Backend Development
Web Frontend Development
Authentication & Authorization
Test-Driven Development
Open-Source Development
Version Control (Git)

Cloud & Infrastructure

AWS Infrastructure as Code
Terraform Docker
Container Orchestration
PostgreSQL Relational Databases
NoSQL Databases

Scientific Computing

Signal Processing Machine Learning
Numerical Methods Data Analysis

LANGUAGES

Portuguese (Native)
English (Fluent)
French (Intermediate)



- Contributed to the maintenance and upgrade of the Tile Muon Digitizer Board software during the **Long Shutdown 2** (LS2), ensuring compatibility with upgraded detector electronics.
- Returned on-site in 2021 for a short mission to design and implement a **machine learning**-based energy estimation tool using **C++** and **Python**, fully integrated into Athena workflows.
- Worked remotely as a **Data Quality Validator**, analyzing detector data integrity and software reconstruction outputs for the Tile Calorimeter.
- Designed and developed an open-source pulse generator and signal simulation library in **C++** with **Python** bindings, enabling realistic modeling of calorimeter electronics for physics studies and algorithm development; the tool has been adopted by multiple international research groups.
- Contributed from 2023 to 2024 to an authorship qualification task focused on **linear** and **machine-learning**-based energy reconstruction algorithms for the upgraded Tile Calorimeter electronics targeting the **High-Luminosity LHC** (HL-LHC).
- Achieved **full authorship** status within the ATLAS Collaboration in 2024.

Software Engineering Consultant

Prime IT

📅 Feb 2022 – Present

📍 Remote – Brazil

- Working as a software engineering consultant through Prime IT, currently assigned to **Siteimprove**.

Technical Lead & Senior Full-Stack Software Engineer

O2 Filmes

📅 Sept 2018 – Jan 2022

📍 Remote – Brazil

- Led the design and development of a collaborative online text editor and project management platform dedicated to screenwriters.
- Architected and implemented back-end services using **Ruby on Rails**, **MongoDB**, and **Node.js**, and front-end applications using **React** and **Etherpad**, following a **microservices**-based, cloud-native architecture.
- Designed infrastructure as code using **Terraform**, deploying containerized services with **Docker** on **Amazon ECS** clusters.
- Took ownership of the full software lifecycle, including system design, implementation, automated testing, deployment pipelines, and operational monitoring.
- Acted as technical lead by mentoring developers, defining architectural decisions, prioritizing backlog items, and coordinating development iterations and A/B experiments.
- Implemented advanced features such as real-time multi-user collaboration, data visualization with **D3.js**, notification systems, file import/export pipelines, and complex UI interactions.

Full-Stack Software Engineer

DataHex Computer Technology

📅 May 2016 – Jun 2018

📍 Nova Friburgo, Brazil

- Designed and developed an event ticketing application for **Android** using **Java**, including data synchronization and Bluetooth-based receipt printing.
- Designed and implemented a **cloud-based** Point of Sale system with business management features, using **Node.js** for back-end services and **AngularJS** for front-end applications, following a **microservices** architecture.
- Developed a cross-platform desktop application using **TypeScript** and **Electron**, supporting data synchronization over the internet and across multiple local network nodes using **peer-to-peer** communication.
- Implemented a centralized authentication and authorization system based on **OAuth 2.0** and **OpenID Connect**.

REFEREES

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✉ Nova Friburgo, Brazil

ACHIEVEMENTS



Breakthrough Prize in Fundamental Physics – Breakthrough Prize Foundation

Awarded collectively to all members of the ATLAS Collaboration at CERN for their contributions to the discovery of the Higgs boson and to major advances in particle physics.



ATLAS Authorship Qualification – CERN

Achieved full authorship status in the ATLAS Collaboration after completing a one-year qualification project focused on software development, detector performance, and data quality activities.



Cum Laude Honors – Rio de Janeiro State University

Graduated with academic distinction in recognition of sustained high academic performance.



DELF B2 Certification – République Française

French language diploma issued by the French Ministry of National Education, certifying proficiency at the B2 level.

Full-Stack Software Engineer & Mobile Developer

Vista Group Network

📅 Jun 2014 – Apr 2016

📍 Nova Friburgo, Brazil

- Developed a **cloud-managed** parking management system using **Android** devices for ticketing and infringement monitoring, with mobile Android application written in **Java** and back-end services in **.NET**.
- Developed cross-platform **iOS** and **Android** mobile applications for parking ticket purchases using **Ionic** and **JavaScript**.
- Designed and implemented a computer vision solution to detect free parking spaces using surveillance camera feeds, developed in **Python** with **OpenCV**.

OPEN SOURCE CONTRIBUTIONS

Athena

The ATLAS Experiment's main offline software

🔗 https://gitlab.cern.ch/atlas/athena/-/merge_requests?state=all&author_username=ginaciog

Lorenzetti

A general-purpose framework for signal reconstruction and triggering in high-energy calorimetry

🔗 <https://github.com/lorenzetti-hep/lorenzetti>

Calorimetry Pulse Simulator

A pulse generator and signal simulation library for calorimeter electronics, with C++ core and Python bindings

🔗 <https://github.com/ingoncalves/calorimetry-pulse-simulator>

Etherpad

A real-time collaborative editor for the web

🔗 <https://github.com/ether/etherpad-lite/commits?author=ingoncalves>

Scilab

Free and Open Source software for numerical computation

🔗 <https://github.com/scilab/scilab/commits?author=ingoncalves>